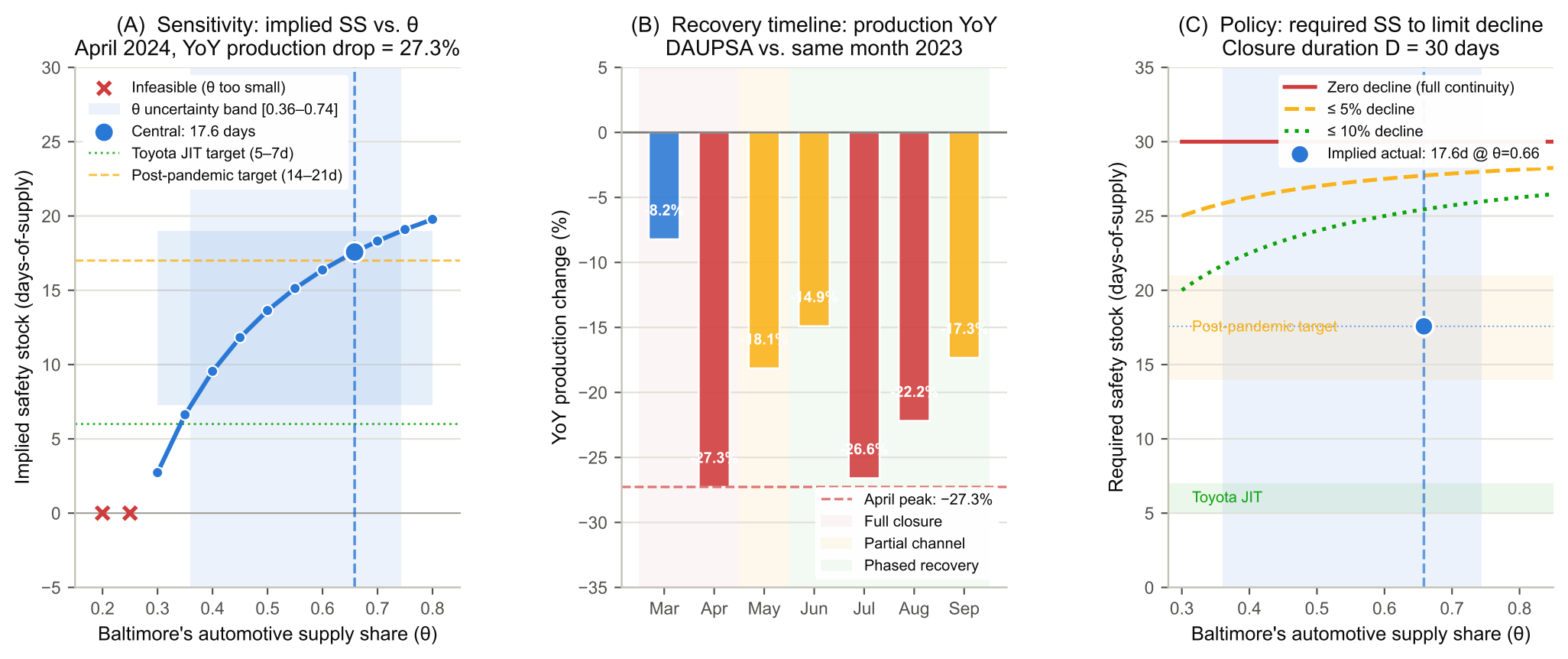
(A) Sensitivity: implied SS vs. θ
April 2024, YoY production drop = 27.3%
Infeasible (θ too small)
θ uncertainty band [0.36–0.74]
Central: 17.6 days
Toyota JIT target (5–7d)
Post-pandemic target (14–21d)
Implied safety stock (days-of-supply)
Baltimore's automotive supply share (θ)
(B) Recovery timeline: production YoY
DAUPSA vs. same month 2023
YoY production change (%)
-8.2%
-27.3%
-18.1%
-14.9%
-26.6%
-22.2%
-17.3%
April peak: −27.3%
Full closure
Partial channel
Phased recovery
Mar
Apr
May
Jun
Jul
Aug
Sep
(C) Policy: required SS to limit decline
Closure duration D = 30 days
Zero decline (full continuity)
≤ 5% decline
≤ 10% decline
Implied actual: 17.6d @ θ=0.66
Post-pandemic target
Toyota JIT
Required safety stock (days-of-supply)
Baltimore's automotive supply share (θ)